

SOFTWARE ENGINEERING LAB – 2

(UE24CS341A)

Agile Backlog Creation & Sprint Simulation in Jira

Team Number: 007

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Problem Statement 7: Hostel Maintenance & Issue Ticketing System

EPIC: Epic 2 – Ticket Triage & Staff Dispatch (FR-002 and NFR-001)

Note: This report covers only Epic 2, my individual contribution from the Lab 1 report. The remaining requirements are covered separately by my teammates: Epic 1 – FR-001 & FR-005: P. Saanvi; Epic 3 – FR-003: Kaustubh Akash; Epic 4 – FR-004 & NFR-002: Kasissnu Ssinha.

Lab Objective

This lab converts the requirements FR-002 and NFR-001 from the Lab 1 report into an Agile Epic and User Stories in Jira, estimates effort using the Fibonacci scale, prioritises the backlog, simulates two one-week sprints on a Scrum board, and tracks sprint progress using the Burndown Chart.

1. Epic (Derived from FR-002 and NFR-001)

The two requirements are grouped into one Epic: FR-002 defines how the Maintenance Warden assigns open tickets to specific technicians from an active staff roster, while NFR-001 enforces system performance by triggering automated SMS/Email escalation alerts whenever high-priority tickets exceed a 24-hour SLA without progress.

Epic Key	Epic Name	Mapped Requirements	Description
HMT-1	Ticket Triage & Staff Dispatch	FR-002 and NFR-001	Empower wardens to assign maintenance staff based on availability and workload, and automatically escalate unresolved high-priority tickets exceeding 24h SLA.

2. User Stories for Epic 2

The Epic is broken into 5 User Stories using the format: As a [role], I want [goal], so that [benefit].

Key	Story	As a...	I want...	So that...	Covers
HMT-2	2.1 Staff Dropdown Assignment	Maintenance Warden	to select and assign a staff technician from an active staff dropdown	responsibility for repair is explicitly designated	FR-002
HMT-3	2.2 Technician Workload Indicator	Maintenance Warden	to view the count of open tickets currently assigned to each technician	work is distributed equitably across the maintenance crew	FR-002

HMT-4	2.3 Dispatch Push Notification	Maintenance Staff	to receive an automated SMS/ push notification upon receiving an assignment	I can inspect and attend to the ticket immediately	FR-002
HMT-5	2.4 Automated 24-Hour SLA Escalation	Maintenance Warden	the system to automatically send SMS/Email alerts when high-priority tickets exceed 24 hours without status change	critical residential emergencies are never overlooked	NFR-001
HMT-6	2.5 Ticket Reassignment Workflow	Maintenance Warden	to reassign a ticket to a substitute technician if the initial assignee is unavailable	repairs are not delayed by staff absence	FR-002

Acceptance criteria were taken from Lab 1. FR-002: Warden selects staff from dropdown, and ticket status transitions to 'Assigned' with assignee logged in the audit trail. NFR-001: Automated background daemon queries tickets every 15 minutes; all high-priority tickets unprogressed past 24 hours trigger SMS/Email alerts to warden within 60 seconds.

3. Jira Backlog – Epic, User Stories and Story Points

The screenshot shows Epic 2 with all 5 User Stories, the priority icon and story points on each row, and the Epic panel reporting 5 work items and an estimate of 24 points.

The screenshot displays the Jira Backlog for the 'Hostel Maintenance & Issue Ticketing System' (HMITS board). The 'Backlog' tab is selected, showing a list of 5 work items under the epic 'Ticket Triage & Staff Dispatch' (HMTS-1). The epic summary on the left indicates 5 work items, 0 completed, and an estimate of 24 points. The backlog list shows the following items:

Item	Key	Summary	Status	Priority	Story Points
1	HMTS-2	Staff Dropdown Assignment	To Do	High	5
2	HMTS-3	Technician Workload	To Do	High	5
3	HMTS-4	Dispatch Push Notification	To Do	Medium	3
4	HMTS-5	Automated 24-Hour SLA Escalation	To Do	High	8
5	HMTS-6	Ticket Reassignment	To Do	Medium	3

Figure 1: Jira Backlog – Epic 2 (HMT-8) with stories (5, 5, 3, 8, 3 = 24 pts).

4. Backlog Prioritization

User Stories were ordered by business value and dependency. The backlog order in Jira is 2.1, 2.2, 2.3, 2.4, 2.5.

Key	Story	Priority	Story Points
HMT-2	2.1 Staff Dropdown Assignment	High	5
HMT-3	2.2 Technician Workload Indicator	High	5
HMT-4	2.3 Dispatch Push Notification	Medium	3
HMT-5	2.4 Automated 24-Hour SLA Escalation	High	8
HMT-6	2.5 Ticket Reassignment Workflow	Medium	3

- 2.1 and 2.2 (High): Core dispatch functionality; without assignment mechanics and workload counts, dispatch is blind and uncoordinated.
- 2.3 (Medium): Notification integration that bridges warden assignment to the field technician.
- 2.4 (High): Critical SLA enforcement daemon protecting against safety and facility violations.
- 2.5 (Medium): Operational fallback allowing workload rebalancing when personnel take leave.

Total Story Points (Epic 2): 24 | High: 18 | Medium: 6

5. Story Point Estimation – Planning Poker

Story points were estimated independently by each team member using the Fibonacci scale (1, 2, 3, 5, 8, 13...). The individual estimates were compared, differences were discussed, and the team agreed on the final estimates.

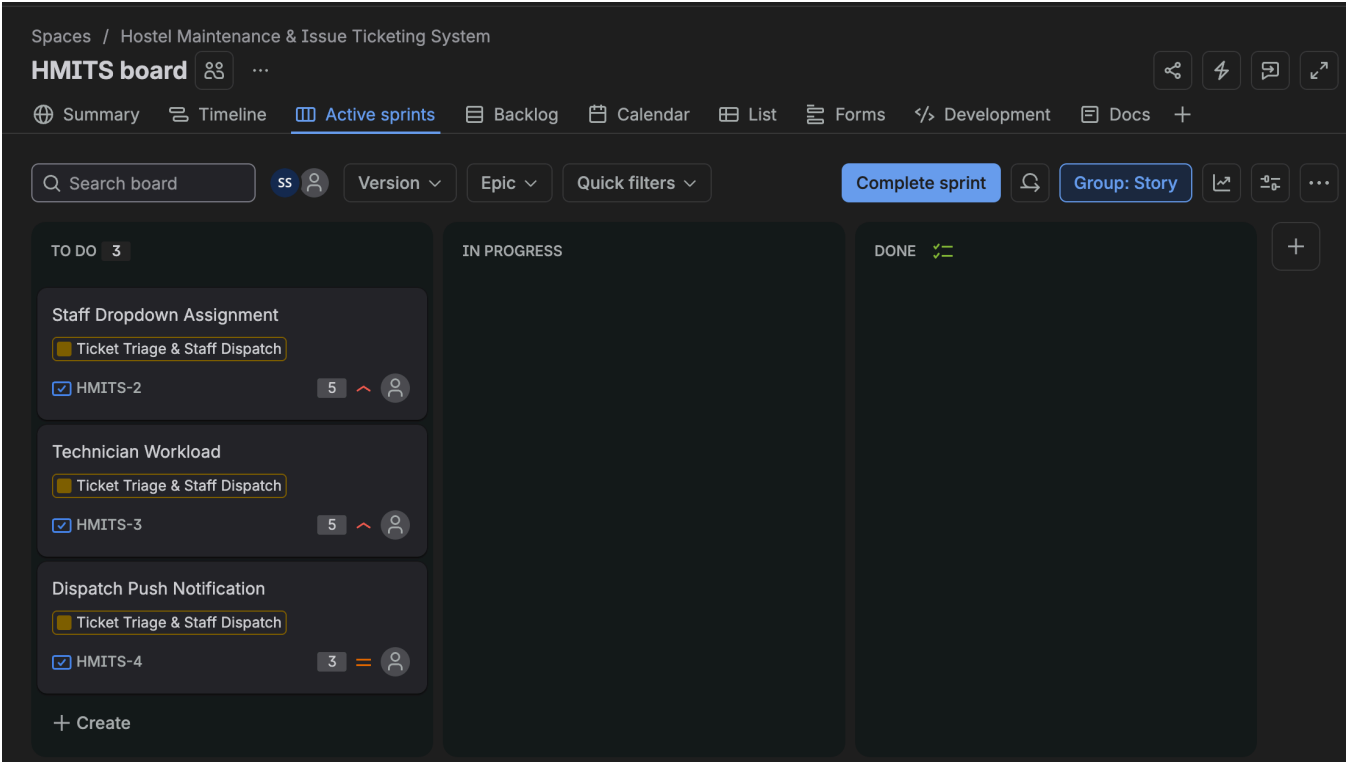
Story ID	Story Title	Saanvi	Sonia	Kaustubh	Kasisnu	Agreed
HMT-2 / 2.1	Staff Dropdown Assignment	5	5	5	3	5
HMT-3 / 2.2	Technician Workload Indicator	5	5	3	5	5
HMT-4 / 2.3	Dispatch Push Notification	3	3	3	5	3
HMT-5 / 2.4	Automated 24-Hour SLA Escalation	8	8	8	8	8
HMT-6 / 2.5	Ticket Reassignment Workflow	3	3	5	3	3

6. Sprint Planning & Simulation

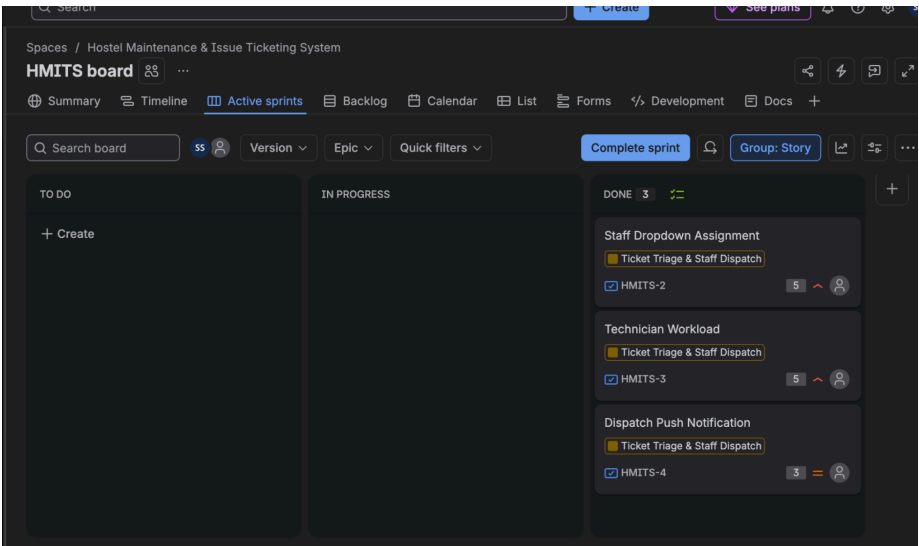
Sprint 1 (Duration: 1 week)

Sprint Goal: Establish core warden dispatch capabilities, technician workload tracking, and instant alerts.

Key	Story	Priority	Pts	Status
HMT-2	2.1 Staff Dropdown Assignment	High	5	Done
HMT-3	2.2 Technician Workload Indicator	High	5	Done
HMT-4	2.3 Dispatch Push Notification	Medium	3	Done



Sprint 1 Velocity: 13 pts committed | 13 pts completed | 0 pts rolled over



Sprint 2 (Duration: 1 week)

Sprint Goal: Implement background SLA monitoring daemon and ticket reassignment exception workflows.

Key	Story	Priority	Pts	Status
HMT-5	2.4 Automated 24-Hour SLA Escalation	High	8	Done
HMT-6	2.5 Ticket Reassignment Workflow	Medium	3	Done

Sprint 2 Velocity: 11 pts committed | 11 pts completed | 0 pts rolled over

Spaces / Hostel Maintenance & Issue Ticketing System

HMITS board

Summary

Timeline

Active sprints

Backlog

Calendar

List

Forms

Development

Docs

Search board

SS

Version

Epic

Quick filters

Complete sprint

Group: Story

TO DO 2

Automated 24-Hour SLA Escalation

Ticket Triage & Staff Dispatch

HMITS-5

8

Ticket Reassignment

Ticket Triage & Staff Dispatch

HMITS-6

3

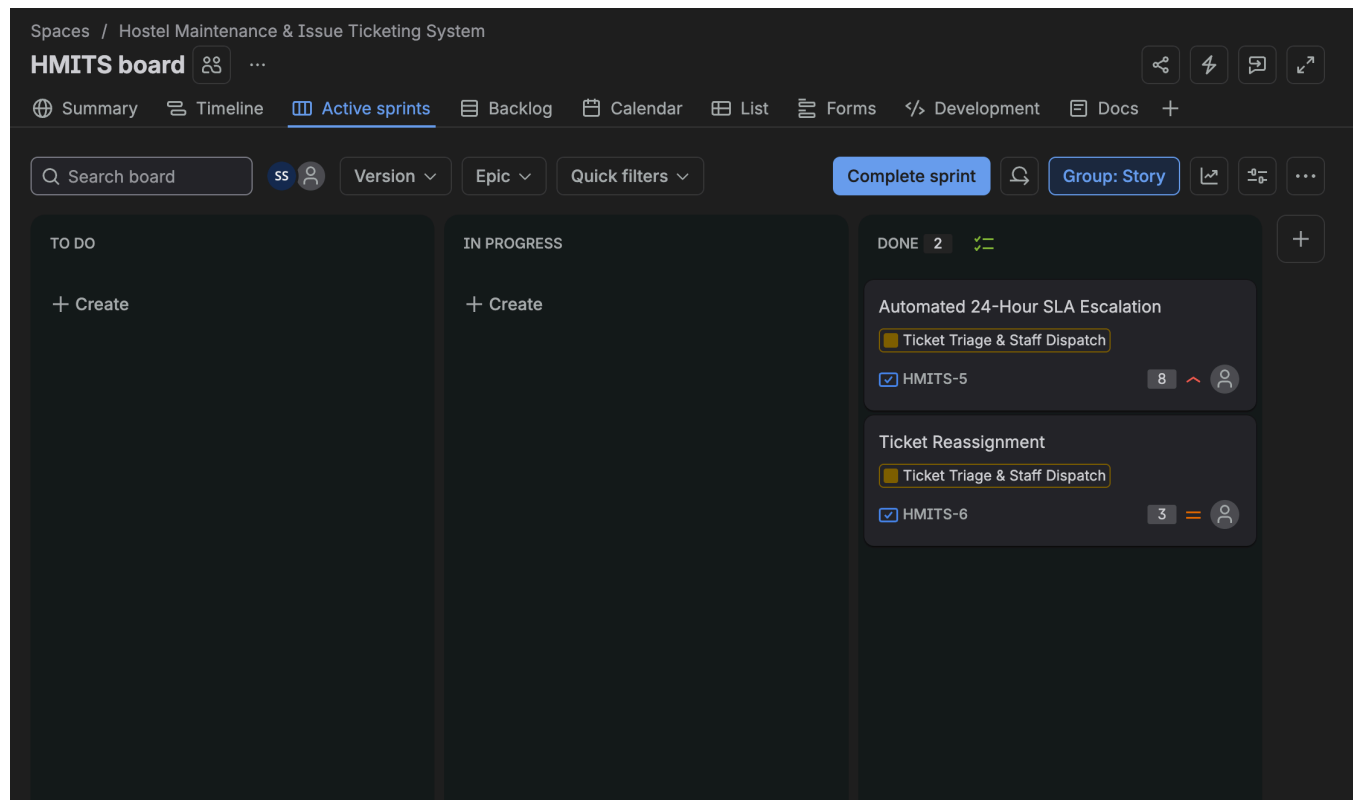
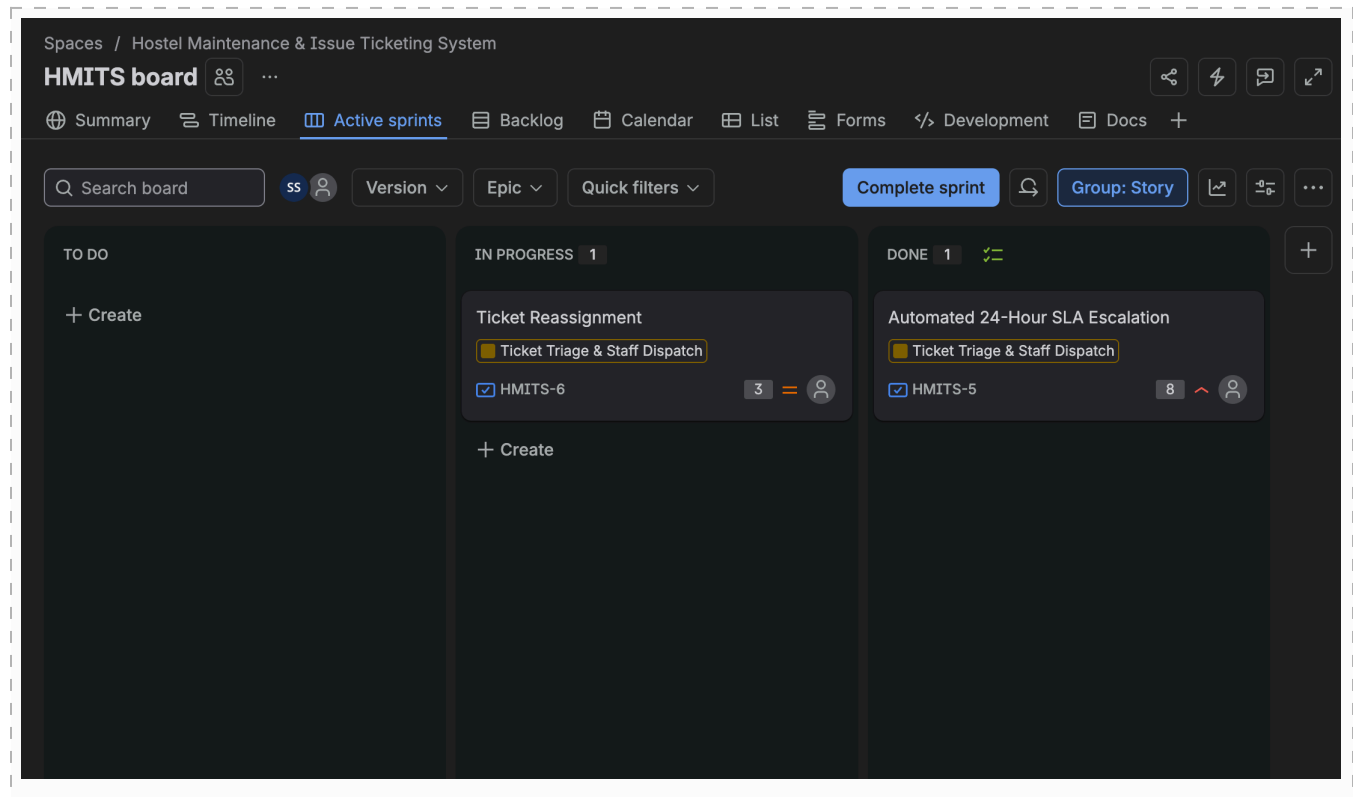
+ Create

IN PROGRESS

DONE

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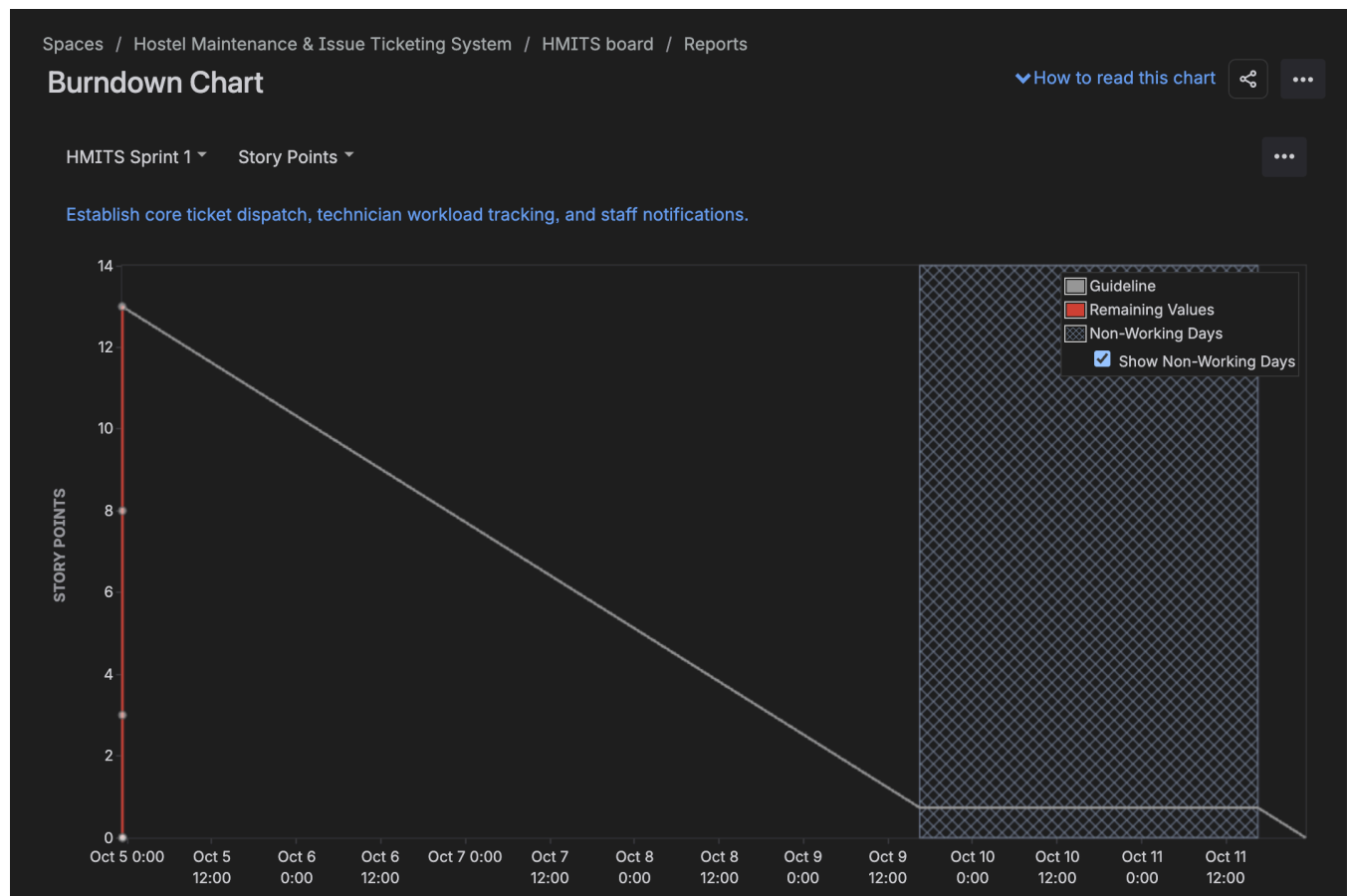
Page 5

Sprint 2 outcome

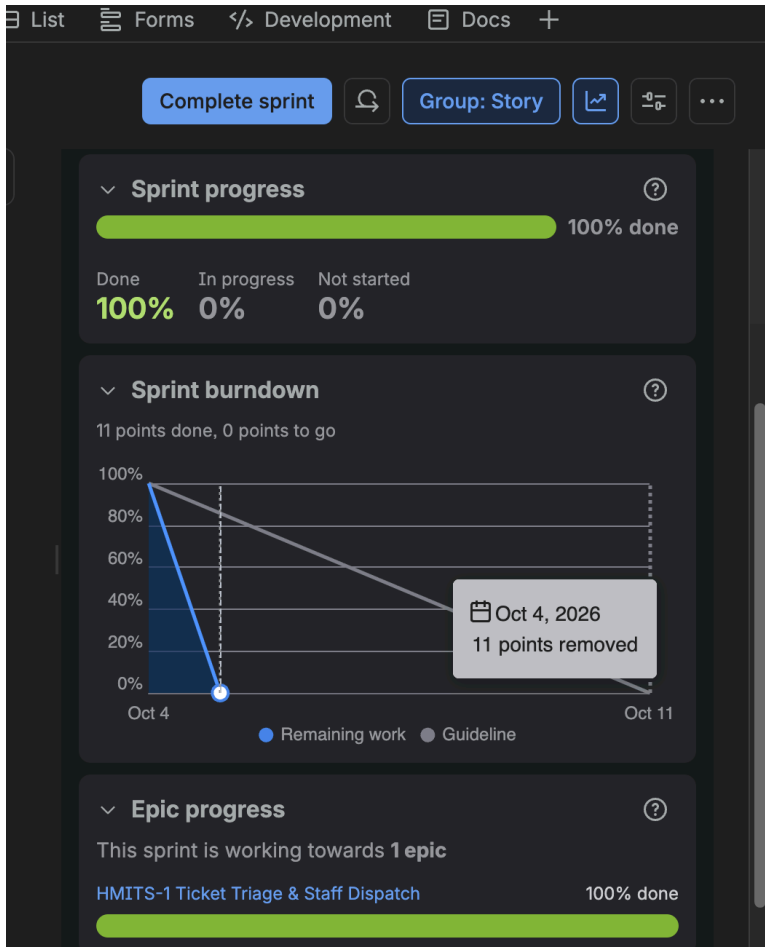
7. Burndown Chart Analysis

The Jira Burndown Chart plots story points remaining (red line) against the ideal guideline (grey line). Each sprint is shown twice: once with non-working days displayed (hatched) and once with them hidden, which makes the guideline easier to read. The event table under each chart lists the exact completion times.

Sprint 1 Burndown (13 pts)



Sprint 2 Burndown (11 pts)



Summary of the burndown data (from the Jira event tables)

Sprint	Sprint	Story Points Completed	Story Points Remaining	Completion	Burndown Observation
Sprint 1	13	13	0	100%	Work dropped from 13 points to 0, with all Sprint 1 work completed by the end of the sprint.
Sprint 2	11	11	0	100%	All 11 points were removed on Oct 4, 2026, resulting in 0 points remaining and 100% completion.
Overall	24	24	0	100%	All planned work across both sprints was completed.

8. Reflection Questions

1. Did your estimations reflect the actual effort?

Yes, the story point estimations reasonably reflected the relative effort and complexity of the work. The simpler stories such as Dispatch Push Notification and Ticket Reassignment were estimated at 3 points, while Staff Dropdown Assignment and Technician Workload were estimated at 5 points. The Automated 24-Hour SLA Escalation story was estimated at 8 points because it involved automated monitoring and escalation logic and had greater complexity and uncertainty. Overall, the estimates provided a reasonable representation of the relative effort required.

2. Was your backlog well-prioritized?

Yes, the backlog was prioritized according to the importance and urgency of the maintenance workflow. Staff Dropdown Assignment and Technician Workload were given high priority because they support the core ticket dispatch process. Automated 24-Hour SLA Escalation was also given high priority because unresolved high-priority tickets require escalation. Dispatch Push Notification and Ticket Reassignment were given medium priority. This ordering allowed the core dispatch functionality to be addressed before the additional exception-handling functionality.

3. How did your simulated sprint align with your plan?

The simulated sprints aligned with the planned work distribution. Sprint 1 contained Staff Dropdown Assignment, Technician Workload, and Dispatch Push Notification, totaling 13 story points. These stories were moved from To Do through In Progress to Done. Sprint 2 contained Automated 24-Hour SLA Escalation and Ticket Reassignment, totaling 11 story points, and both were also completed. Therefore, all 24 story points planned across the two sprints were completed successfully.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart showed that the planned work was completed within the sprint. In Sprint 2, all 11 story points were completed, resulting in 0 points remaining and 100% sprint completion. The sharp decrease in remaining work indicates that the team was able to complete the committed work within the planned sprint period. The chart also provides a useful indication of the team's capacity and can help improve future sprint planning and estimation.